



Voting System Anomaly Root Cause Analysis Template v2.0

Root Cause Analysis for:

VV40ECT-234: VxSCAN LIGHTNING SURGE, DISPLAY STOPPED SHOWING BALLOT ON SCREEN - REV. 1
VxSUITE, VERSION 4.0 AND EAC CERTIFICATION #VXS4

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Introduction

This RCA documents the investigation into an anomaly during standard electrical testing of VxScan, where the test software stopped showing the scanned ballots visually on screen. This happened during tests of resistance to lightning surge, where test software keeps VxScan in a state of automatic continuous operation, or “shoeshine” mode. Potentially it could occur in any test using this shoeshine mode, because it was linked to rescanning the same single ballot hundreds of times and getting dirty. Investigation was important to (1) check that the test software was continuing to scan the ballots even if not displaying them on screen, and (2) understand why an especially dirty ballot led to lack of displaying the ballot, particularly since this was not an issue for previous electrical tests in pre-certification testing.

Anomaly Description

Complete all sections. Descriptions must be as detailed as possible, while being clear and concise since the anomaly is the source of the entire RCA. This detail should include a complete list and/or description of the “symptoms” of the anomaly and the conditions present which the symptoms occurred.

<u>Date of Anomaly:</u> April 8, 2026	<u>Time of Anomaly:</u> 10:04am
<u>Place of Anomaly:</u> Element, Longmont, CO	<u>Person identifying Anomaly:</u> Jessica Myers, VotingWorks
<u>Expected Results of actions leading up to anomaly:</u> VxScan test software (Hardware Test Utility) displays all scanned ballots in shoeshine mode over time, even if rescanned many times and dirty.	

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Detailed description of the event / anomaly:

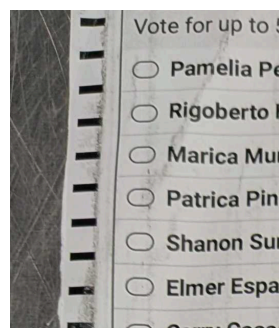
VxScan unit SC-11-003 was undergoing standard Lightning Surge testing (IEC 61000-4-5) with SLI at Element in Longmont, CO. With the VxScan in shoeshine mode, black smearing was occurring on the ballot. The ballot was replaced with a new one. Black smearing occurred with this second ballot. Operator error at this point caused the need for a reboot, and then the ballot was replaced. Smearing occurred again. Then the display stopped showing the visual scan of the ballot on screen. The rest of the test software was functioning as expected. Logs were pulled for analysis to check that images were still being saved and analyzed, and the test continued to completion. Logs were again pulled at the end for more analysis.

If the anomaly is repeatable, provide step by step instructions to recreate it:

1. Set up the VxScan unit with the Hardware Test Utility in shoeshine mode.
2. Insert a test ballot sheet of paper, and let the single ballot rescan up to hundreds of times.
3. Observe the quality of the single ballot and the display over time (potentially hundreds of scans), to see if the scan image is updating. Replace the ballot if needed and continue to observe.

Chronology of Events / Timeline

Provide a detailed chronology of the events leading up to, and following, the anomaly. Add additional events if necessary.

ID	Date/Time	Description	Entity Org/person	Result / Notes
1	9/3/25, 8:05am Mountain Time	VxScan begins lightning surge tests with SLI at Element in Longmont, CO.	SLI, and Jessica Myers, Jesse DeWald, Arsalan Sufi, Brian Donovan, VotingWorks	VxScan unit SC-15-003 is set up for lightning surge testing without issue. As testing proceeds, it is noted that the single ballot shows streaking and degradation over time as it is scanned repeatedly (example shown). 

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				It was reported to occur not just in the lightning surge test. Ballots are replaced as needed, and the VxScan unit is cleaned according to normal protocols between them. A second attempt to replace the ballot leads to operator error requiring a restart of the VxScan. Logs are pulled for analysis. Offsite, VotingWorks staff reproduce and analyze the issue, confirming that very dirty ballots with markings along the timing marks will scan in the test software, but will not show scans on the screen. This can cause some confusion during testing but does not prevent underlying scanning of images. Plans were made to confirm the images were actually scanned by analyzing the full logs following test completion.
2	9/3/25, 2:48pm	Lightning surge tests conclude with other issues, while VotingWorks extracts scanned images from test software.	SLI, and Jessica Myers, Arsalan Sufi, VotingWorks	The test concludes with no other issues. Logs are analyzed to check that images were saved throughout the test. VotingWorks extracted the images from the system for full review by SLI, confirming all scans were present even not displaying them on screen in the test utility software.

Investigative Team and Method

This section shall describe how the investigative team is assembled by the voting system manufacturer, who it consists of, and how it gathers the data to be used in the analysis. Include the RCA method employed by the manufacturer in conducting the analysis and why this method was used.

Names and Positions of members of the investigation team:
Jessica Myers, Head of Compliance
Jesse DeWald, Head of Hardware
Arsalan Sufi, Head of Software
Brian Donovan, Software Engineer

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Describe the data gathering process:

Jessica Myers and Jesse DeWald initiated the investigation as the VotingWorks staff on-site at Element during these tests. They received initial reports from SLI/Element about the ballot streaking issue and subsequent lack of displaying scans onscreen in the Hardware Test Utility, and coordinated getting logs of the issue.

Arsalan Sufi and Brian Donovan analyzed the underlying test software behavior that could lead to the display issue. They confirmed the issue was not present in the official production software, and found it was rather a limitation of the Hardware Test Utility not handling especially dirty ballots. Arsalan coordinated extracting all the images from the test unit so that SLI could conduct final analysis to confirm they all were present as expected.

Describe which methodology(s) is used to conduct the root cause analysis:

A small fault tree analysis guided the investigation. Early evidence pointed to either:

- 1) Physical causes related to rescanning very dirty paper
- 2) Software causes related to scanning very dirty paper

Within these branches, the physical causes were deemed less likely after properly cleaning the unit and replacing the paper, and investigating print quality of the test ballots.

The software causes were deemed more likely, since it was known that disrupted timing marks will prevent interpretation of a scanned ballot. It would not have disrupted scanning itself, but it could have caused other downstream effects leading to the lack of displaying the ballot. Within this branch, possible explanations for the observed behaviors were analyzed, including:

- 1) Timing mark detection errors are just not updating on-screen
- 2) Images are not showing up on screen after timing mark errors

Reproduction of the issue separately confirmed the final root cause of images just not displaying on screen after timing mark errors, despite still being saved and analyzed by the software.

Findings and Root Cause

Describe the findings of the investigation and explain the root cause(s) based on these findings. If the RCA results in findings that are not directly related to the root cause of the anomaly, these should also be captured as manufacturer product/process improvement steps in an effort to improve the voting system.

The root cause of this issue was that the Hardware Test Utility was not built to show images for scans that are unsuccessfully interpreted. In this case, the repeatedly scanned single ballot had markings that made the timing marks uninterpretable, and so the test software would scan the ballots, be unable to interpret them, and not display them due to that interpretation error. Logging and reproduction of the issue at VotingWorks confirmed this.

A secondary but important note is that the VxScan system still resists lightning surge according to standard requirements. The lightning surge test does not cause this issue, but rather the use of shoeshine mode to rescan a ballot many times in the Hardware Test Utility could cause this more generally, due to the ballot becoming very dirty. Rescanning the same ballot hundreds of times would not happen in real-life election scenarios. Furthermore, the official product software app handles scanned ballots more robustly than this test software, because the real election software never displays scanned ballots on-screen.

Corrective Action(s)

No corrective action is needed for the official production software for VxScan, because the official software never rescans the same ballots repeatedly and never displays scanned ballots. The observed anomaly here is specific to the Hardware Test Utility used for shoeshine mode during EMC testing.

However, to improve ease of testing in the future, the Hardware Test Utility was updated to handle timing mark interpretation errors better and still display scans of degraded paper. Additionally, the test software was patched to improve handling the replacement of paper after shoeshine mode sessions. Both these updates would reduce confusion reported in this anomaly. These code updates can be seen in the open source pull request here:

- <https://github.com/votingworks/vxsuite/pull/8269>

Solution Management

The purpose of this section is to manage the corrective action(s) moving forward. This should detail all process changes to manage those corrective actions, and steps taken to ensure the actions eliminate the anomaly over time.

The VxScan Hardware Test Utility software was updated, tested, and reviewed, according to standard development practices. It will be incorporated into future EMC testing requiring use of shoeshine mode.